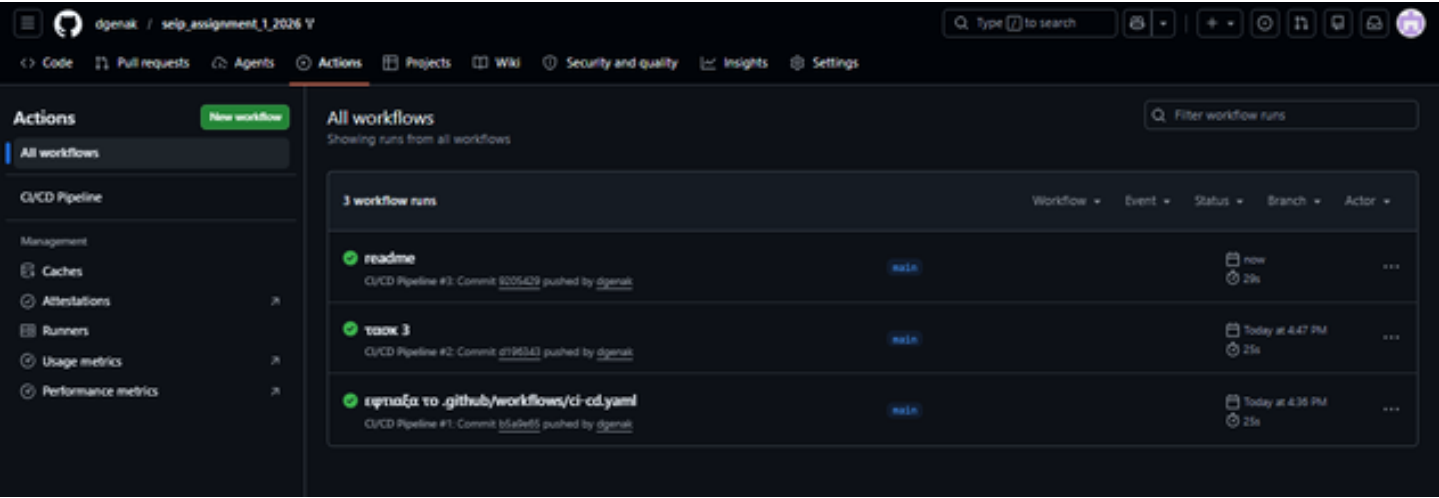


GitHub Link: [https://github.com/dgenak/seip\\_assignment\\_1\\_2026](https://github.com/dgenak/seip_assignment_1_2026)

SCREENSHOTS

CI/CD Pipeline Runs



Kubernetes Cluster Status

```
PS C:\Users\dim> kubectl get all -n default
NAME                                READY    STATUS    RESTARTS    AGE
pod/echo-api-698f78f7ff-7nr55      1/1     Running   0            2m13s
pod/echo-api-698f78f7ff-hnnbm      1/1     Running   0            2m13s
pod/echo-api-698f78f7ff-wrqp4      1/1     Running   0            2m14s

NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
service/echo-api-service           ClusterIP      10.105.173.80 <none>         80/TCP      2m14s
service/kubernetes                  ClusterIP      10.96.0.1     <none>         443/TCP     26m

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/echo-api            3/3      3              3            2m14s

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/echo-api-698f78f7ff 3          3          3        2m14s
PS C:\Users\dim> |
```

## ConfigMap and Secret Verification

```
PS C:\Users\dim> kubectl get configmap,secret
NAME                                DATA  AGE
configmap/app-config                2      75m
configmap/kube-root-ca.crt          1      100m

NAME                                TYPE    DATA  AGE
secret/app-secret                   Opaque  1      75m
PS C:\Users\dim>
```

## Greeting Endpoint Response

```
PS C:\Users\dim> curl http://localhost:8080/

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is
parsed.
RECOMMENDED ACTION:
Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): y

StatusCode      : 200
StatusDescription : OK
Content          : {"message":"Welcome to the Software Engineering in Practice Assignment
Cluster!","environment":"production"}
RawContent       : HTTP/1.1 200 OK
                  Connection: keep-alive
                  Keep-Alive: timeout=5
                  Content-Length: 108
                  Content-Type: application/json; charset=utf-8
                  Date: Fri, 22 May 2026 15:03:37 GMT
                  ETag: W/"6c-sLe+zwZH8jv0Y2sDz+X...
Forms            : {}
Headers          : {[Connection, keep-alive], [Keep-Alive, timeout=5], [Content-Length, 108], [Content-Type,
                  application/json; charset=utf-8]...}
Images           : {}
InputFields      : {}
Links            : {}
ParsedHtml       : mshtml.HTMLDocumentClass
RawContentLength : 108

PS C:\Users\dim> |
```

## Secure Config Endpoint Response

```
PS C:\Users\dim> curl http://localhost:8080/secure-config

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is
parsed.
    RECOMMENDED ACTION:
    Use the -UseBasicParsing switch to avoid script code execution.

    Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): y

StatusCode      : 200
StatusDescription : OK
Content         : {"status":"Authorized","injected_secret_suffix":"*****2345"}
RawContent      : HTTP/1.1 200 OK
                  Connection: keep-alive
                  Keep-Alive: timeout=5
                  Content-Length: 80
                  Content-Type: application/json; charset=utf-8
                  Date: Fri, 22 May 2026 15:04:06 GMT
                  ETag: W/"50-04Zc000zx3nx1PeWetwM...
Forms           : {}
Headers         : {[Connection, keep-alive], [Keep-Alive, timeout=5], [Content-Length, 80], [Content-Type,
                  application/json; charset=utf-8]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 80
```

## README.md on GitHub

Preview Code | Blame 48 lines (41 loc) · 2.31 KB



## 1. Clone the Repository

To download the project, I executed: `git clone https://github.com/dgenak/seip\_assignment\_1\_2026.git` to enter the folder: `cd seip_assignment_1_2026`

## Starting Minikube

I started the Kubernetes cluster using the Docker driver: `minikube start --driver=docker`

## Manifests (Sequential)

I applied the Kubernetes manifests in the correct order: `ConfigMap kubectl apply -f k8s/configmap.yaml Secret kubectl apply -f k8s/secret.yaml Deployment kubectl apply -f k8s/deployment.yaml Service kubectl apply -f k8s/service.yaml`

## Checking that the Pods are Running

I waited to see that the 3 pods are running: `kubectl get pods -n default`

## Port Forwarding

I connected the local port 8080 to the service: `kubectl port-forward svc/echo-api-service 8080:80`

## Testing the Endpoints

Greeting Endpoint: `curl http://localhost:8080/` Secure Config Endpoint: `curl http://localhost:8080/secure-config`

2. I used port forwarding to map the ClusterIP service to localhost: `kubectl port-forward svc/echo-api-service 8080:80` This allowed me to access the application at <http://localhost:8080>

## 3. AI Integration

I used AI in three specific areas: First, AI helped me translate the README to English. I needed the technical commands to be accurate and AI did that well. Second, AI helped me with the syntax of the YAML files. For example, I had wrong indentation in the `deployment.yaml` and AI spotted it and fixed it. Third, when I tried to create the `.github/workflows` folder on Windows, the `mkdir` command didn't work. AI told me to use `New-Item` instead and it worked immediately.

## Utility Analysis

AI was most helpful with syntax debugging and YAML formatting. It caught indentation errors that would break the manifests and helped me understand how `ConfigMap` and `Secret` fields need to match the application expectations.

## Friction Points

AI had limitations with the Base64 encoding - it included a newline character that broke the secret. I had to manually test and use the correct `echo` command.

## Future Architectural Outlook

If I had more time, I would add automated testing in the CI pipeline, create a simple monitoring dashboard to track pod health, and implement automated backups for the cluster.

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